

Day 2 :Power Supply Systems Theory

Task 1 : List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

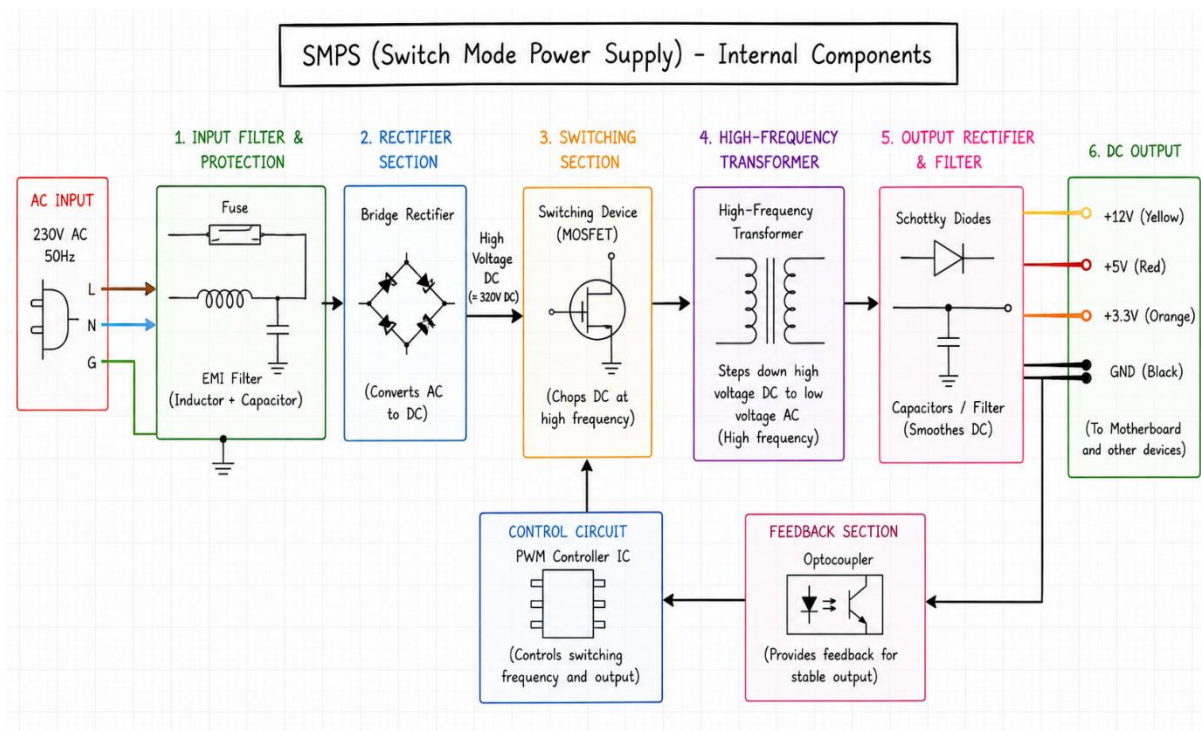
Devices Using SMPS

1. **Desktop Computer** – Uses SMPS to convert AC power into regulated DC power required by the computer components.
2. **LED Monitor** – Uses SMPS to provide stable and efficient DC power to the display and its electronic circuits.
3. **Wi-Fi Router** – Uses an SMPS adapter to convert AC power into the low-voltage DC power required by the router.

Devices Using UPS

1. **Desktop Computer** – Uses a UPS to provide temporary backup power during power cuts and prevent sudden shutdown or data loss.
2. **CCTV Camera System** – Uses a UPS to keep the cameras and recording system working during power interruptions.

Task 2 : Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections



Working of SMPS :

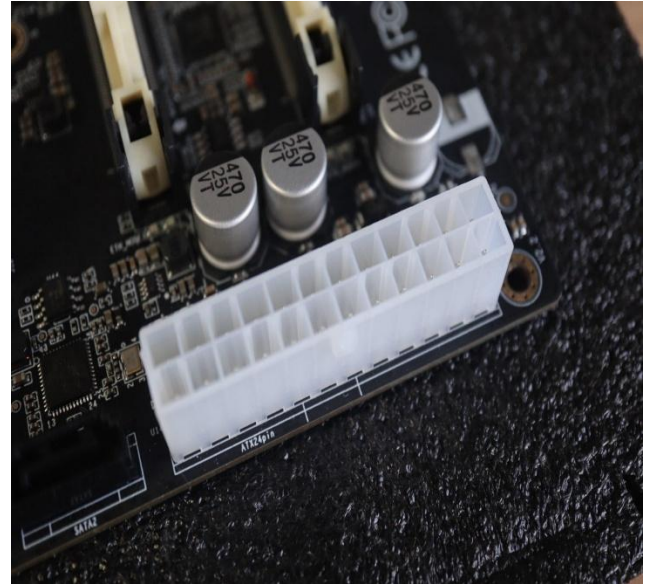
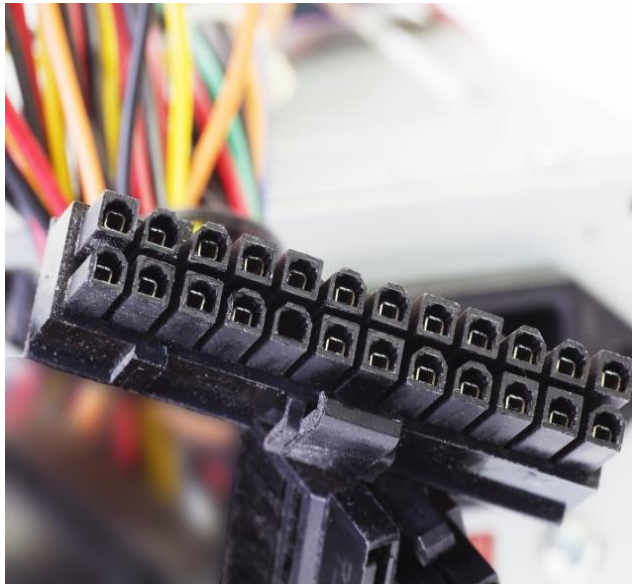
- SMPS (Switch Mode Power Supply) converts **230V AC from the mains supply into low-voltage DC power** required by computer components.
- First, the AC input passes through the **fuse and EMI filter**, which provide protection and reduce electrical noise. Then, the **bridge rectifier** converts AC into high-voltage DC.
- This DC is given to the **MOSFET switching circuit**, which switches the power at high frequency. The high-frequency signal then passes through the **high-frequency transformer**, which reduces the voltage and provides electrical isolation.
- After that, the **output rectifier and capacitors** convert and smooth the signal into stable DC power. Finally, the SMPS provides **+12V, +5V, +3.3V, and GND** outputs to the motherboard and other computer components.
- The **feedback and control circuit** continuously monitors the output and adjusts the switching to keep the voltage stable.

In short:

AC Input → Filter → Rectifier → Switching → Transformer → Output Rectifier & Filter → DC Output

Task 3 : Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Motherboard Connector



Devices powered: Motherboard. It provides the main power connection to the motherboard.

Safety tip: Turn off the PC and unplug the power cable before connecting or disconnecting. Make sure the connector's latch is properly aligned and fully locked.

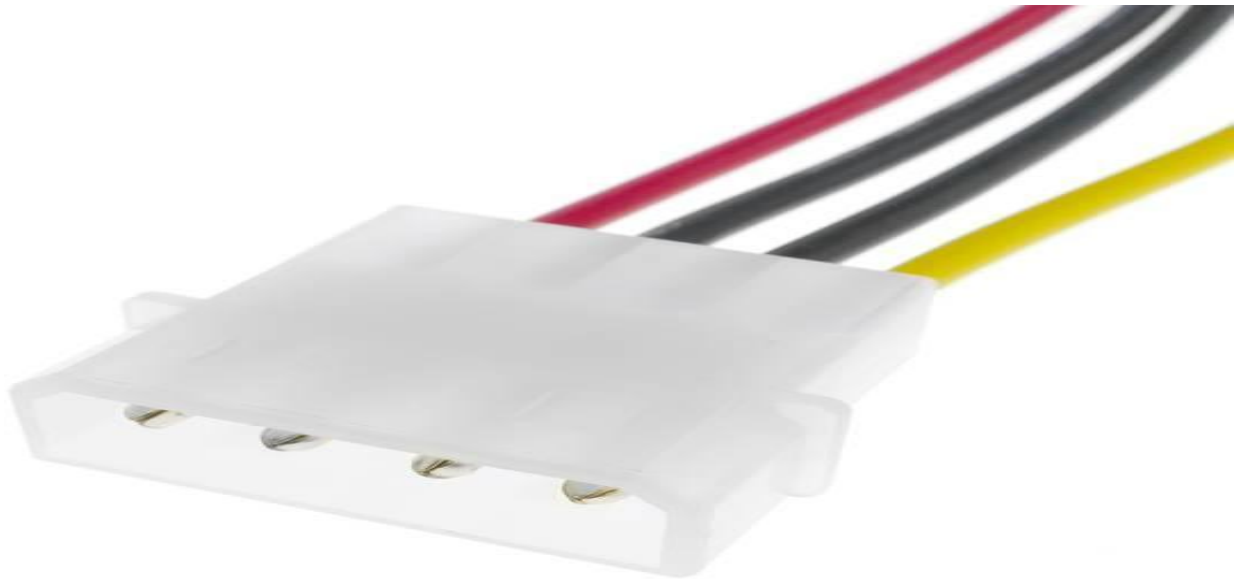
2. SATA Power Connector



Devices powered: SATA HDDs, SSDs, optical drives, and some RGB/fan controllers.

Safety tip: Switch off and unplug the PC before connecting or removing it. Do not force the connector; align its L-shaped key correctly.

3. Molex Connector



Devices powered: Older hard drives, CD/DVD drives, some case fans, and other older peripherals.

Safety tip: Turn off and unplug the PC before removing it. Pull the connector by the plastic plug, not by the wires.

4. PCIe 6/8-Pin Connector



Devices powered: Mainly dedicated graphics cards (GPU) that require additional power. A 6+2-pin connector can be used for either 6-pin or 8-pin PCIe power.

Safety tip: Make sure you use the PCIe cable, not the similar-looking CPU/EPS cable, and push it in until the latch locks.

Step 4 : Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Steps to Connect SMPS to a Gaming PC

1. Turn off and prepare the PC

- Before connecting anything, switch off the SMPS and unplug the power cable.
- Place the PC on a clean, dry surface and make sure all components are properly installed.

2. Connect the motherboard

- Take the **24-pin ATX motherboard power cable** from the SMPS and connect it to the 24-pin power socket on the motherboard.
- Make sure the connector is fully inserted and the locking clip clicks into place.

3. Connect CPU power

- Connect the **4+4-pin CPU/EPS cable** from the SMPS to the CPU power connector, usually located near the top of the motherboard.
- This provides power to the CPU.

4. Connect the graphics card

- If the graphics card requires additional power, use the **PCIe 6-pin, 8-pin, or 6+2-pin PCIe cable** from the SMPS.
- Connect it to the power socket on the GPU until the clip locks.

5. Connect storage devices

- For SATA HDDs and SSDs, use the **SATA power cable** from the SMPS and connect it to the SATA power port on the drive.
- The SATA data cable is connected separately between the drive and motherboard.

6. Check all connections

- Make sure every connector is properly inserted and no cables are loose or touching the fans.
- Do not force any connector into the wrong socket.

7. Turn on the PC

- After checking all connections, connect the SMPS to the wall socket, switch it on, and press the PC power button.
- If everything is connected correctly, the system should start normally.

Cable Summary

Component	SMPS Cable / Connector
Motherboard	24-pin ATX
CPU	4+4-pin CPU/EPS
Graphics Card	PCIe 6-pin / 8-pin / 6+2-pin
SATA HDD/SSD	SATA Power
Older peripherals	4-pin Molex